

Econometrics III
Lindo, Fall 2013
Assignment 6
Due: xxxxx

The usual note applies.

These simulations will take a while to run once they are set up. Please copy and paste the relevant output into your do file (commented out) where appropriate so that the code can be—but need not be—run to see the results of your work.

This exercise has you consider the bias induced by an “Ashenfelter Dip” in a generalized difference in difference model. In particular, the usual such model is:

$$Y_{it} = \alpha_i + \alpha_t + \theta Treat_{it} + \epsilon_{it} \quad (1)$$

where α_i are individual fixed effects, α_t are year fixed effects, and T_{it} is an indicator variable that takes a one when an individual has been treated. (In some scenarios, it may make sense to have this variable take a one in *all* post-treatment periods whereas in others it may make sense to have it “turn back off.”)

PART I: As an intermediate step to considering the consequences of an Ashenfelter Dip, we will begin by considering the consequences of autocorrelated error terms. In order to do so, simulate a model in which there are fifteen periods, $\alpha_i \sim N(20, 4)$, $\alpha_t \sim N(1, 1)$, $\theta = 3$, and $\epsilon_{it} = 0.75u_{it} + 0.25\epsilon_{it-1}$ where $u \sim N(0, 2)$ except in period 1 when $\epsilon_{it} = u_{it}$. Moreover, let a permanent treatment be randomly assigned in period 7 with a probability of 0.5 (such that $Treat_{it} = 1$ for treated individuals in the treatment period *and all subsequent periods*). Run 1000 replications with a panel of 200 individuals in which you retrieve the estimated treatment effect, its standard error estimate, and whether you reject $H_0 : \theta = 3$ at the five-percent level. Do this three times over. First, assume that the errors are iid. Second, allow for heteroskedasticity-robust standard errors. Third, allow the errors to be correlated within individuals over time by clustering. [Note: Run the regressions assuming iid and allowing for clustering using xtreg to speed things up. There’s no way to get heteroskedastic-robust estimates that are not clustered using this xtreg so do that one “longhand.”] What do you learn?

PART II: Now consider a similar setup but instead of having treatment be randomly assigned, it will be based on the outcome that is observed in period 6. In particular, treatment is assigned to all individuals with outcomes below the sample median in this period. Run 500 replications with a panel of 200 individuals, retrieving the estimated treatment effect, its standard error estimate clustered on the individual, and whether the true effect is rejected at the five-percent level. What do you learn?

PART III: One way we test the validity of difference-in-difference estimators is by investigating whether there are “effects” before treatment is actually assigned. Repeat the simulation from Part II but add to the regression model an indicator variable for being “one period before treated” (i.e., a variable that takes a one for the treatment group in period 6). How often do we reject the null hypothesis that the coefficient on this variable is zero? In other words, how often does this “falsification test” successfully diagnose that the model is misspecified?

PART IV: Would controlling for the outcome in period 6 be a useful way of addressing the bias observed in Part II? Explain.

Submit the do-file your TA electronically. If it does not run on our computers unaltered, it will be returned to you for revision. In order to facilitate this, include the following at the top of your code:

```
capture cd "<insert what you want to use as your working directory for this assignment here>"
if c(username)=="apackham" {
    cd "C:\Documents and Settings\apackham\Desktop\ECMT677\Assignments\Assignment 6 DiffInDiffSelection"
}
```